

Richardson's method of segment counting versus box-counting

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Abstract—Fractal analysis has become a popular method in all branches of scientific investigation including biology and medicine. This paper presents solution for many unresolved questions about the methodology of fractal theory, precisely in connection between fractal geometry and fractal analysis. While some concepts in fractal theory are determined descriptively and/or qualitatively, this paper provides their exact mathematical definition or explanation. Also, we present results in applying two basic length-related methods on two dimensional neuronal images and discuss their applicability.

Keywords—Box-counting method; Fractal analysis; Fractal geometry; Image; Neuron; Segment-counting method;

I. INTRODUCTION

Although standard quantitative methods in science are based on classical Euclidean geometry [1], fractal geometry is developed as a new geometry of nature [2-5]. It was conceived in 1975 by Benoît B. Mandelbrot, with aim to describe the complexity of forms and processes met in nature [6,7].

Generally, the *topological dimension* (D_T) of a space or object is defined as the minimum number of coordinates needed to specify a position of any point within it. Thus a straight or irregular line has a dimension 1 because only one coordinate is needed to specify position of a point on the line. Following same deduction, a surface (such as a plane) has a dimension 2, while the inside of a cube, a cylinder or a sphere has dimension 3. The topological dimension D_T is always an integer but D need not be [7].

Early work showed that most commonly biological patterns were characterized by fractal geometry [8-11]. Up to now fractal geometry is being used in diverse research areas [12-14] and is proving to be an increasingly useful tool.

According to the previous conclusions, fractal analysis is derived from fractal geometry [2,3,6]. Today, fractal analysis stands a contemporary nontraditional mathematical method of measuring complexity of patterns in geometry and nature

using the fractal dimension. The complexity and fractal dimension are larger if the object's border is more rugged, branching pattern more abundant and lines more irregular and twister [3,5].

At the time of writing this paper, a search on “fractal theory” (“fractal geometry” or “fractal analysis”) throughout World Wide Web yielded more than 600 articles. In spite of that, no consensus emerged so far about the all-inclusive concept of fractal theory, particularly on fractal analysis and its relation to fractal geometry.

The aim of the present study is to explore concept of fractal throughout fractal geometry and fractal analysis, and to support the hypothesis that fractal geometry, particularly, fractal analysis uses methodologies and concepts of Mandelbrot's fractal geometry and Euclidean traditional geometry. Some related methodological considerations of length-related methods are discussed and their results on two dimensional images from human and rat spinal cord are presented.

II. FRACTAL THEORY

A. Fractal geometry

The basic idea or, the starting point, of fractal geometry is term “fractal” and its definition [7]. Fractals can be *geometrical* and *statistical* [3,15]. Each geometrical fractal should be considered as an infinite ordered set of fractal objects defined on a metric space, with four features [6,10]: (1) the shape of a starting object, (2) recursion (generating, iterated) algorithm, (3) conditions (before all, the property of self-similarity) and (4) fractal dimension. In that case, generators are called *prefractals* [3,4]. The final result of such infinite procedure is the *limit fractal* [7]. The initiator, prefractals and limit fractal represent the geometrical fractal set [15].

To simplify the analysis, we restricted further considerations to plane geometry. In that respect, *geometric figures* are open or closed Euclidean (smooth or polygonal)

lines in a plane. Basic definitions and laws for plane geometry of fractals can be demonstrated on some classical fractal models [6,10]. For that purpose, we choose the *von Koch snowflake figure* (Fig. 1A). The sequential construction of this fractal begins with an equilateral triangle: the initiator (Fig. 1A, left side). The iterative algorithm to generate the von Koch fractal is to repeatedly exchange the middle third of each side of the initiator or preceding figure with two sides of smaller. The result after the first iteration is shown in Fig. 1A (middle side) and that after the second iteration (Fig. 1A, right side).

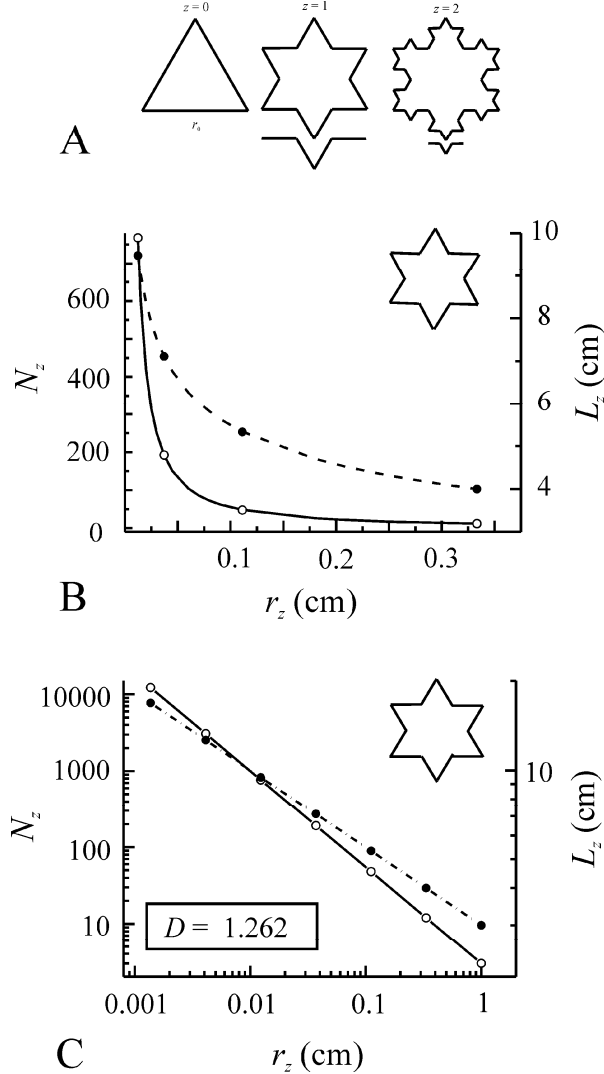


Fig. 1. A - Illustration of an application of iterative algorithms to the constructions of the first (middle) and second-stage generators (right) starting from the equilateral triangle (left) for the von Koch snowflake image. Details below the drawings (middle and right) represent the generating elements for the von Koch prefractals. B - Number of segments N_z (left axes, open circles) and length L_z (right axes, filled circles) of the von Koch prefractals plotted against the segment length r_z on Cartesian axes. The full lines are obtained by fitting the middle of equation (1), and dashed lines, by fitting right side of equation (1) to data points. C - Number of segments and length plotted against the segment length on log-log axes. The straight-line plots are obtained using equation (1).

The length of a segment at the z th stage (r_z), number of segments at the same stage (N_z) and perimeter of that prefractal (L_z) are, respectively,

$$r_z = \frac{r_0}{3^z}, N_z = 3 \cdot 4^z, L_z = 3r_0 \cdot \left(\frac{4}{3}\right)^z. \quad (1)$$

where r_0 is the side length of the initiator (Fig. 1A, left side). From this equation it is seen that the perimeter and number of segments diverge as z approaches infinity. The same holds for the numbers of segments N_z . If the values of N_z and L_z are plotted against the length of segments r_z , the corresponding graphs, fitting the data shown by open and filled circles, exhibit a typical hyperbolic decrease (Fig. 1B).

In establishing any system of geometry, one must start by naming certain notions which cannot be defined analytically and by making certain assumptions (the *axioms*), from which, by the laws of logic, one can develop a consistent system by means of definitions and theorems as deductions from axioms [16]. It seems most likely that fractal and Euclidean geometries rely upon the same statements, except for an axiom, which should be added to Euclidean axioms to establish fractal geometry.

Equation (1) makes the connection between the length of fractal segments and their number at each stage of construction. The qualitative basis of this axiom is our observation [2,17]: the number of segments decreases hyperbolically with the segment length (Fig. 1B). This axiom fulfills the requirement of independency [18] since it cannot be deduced from Euclidean axioms. Equation (1) demonstrates the relations between the three main variables (r_z , N_z and L_z) and the stage of construction (z). Relations between N_z and r_z , as well as between L_z and r_z , being essential for fractal theory, can be considered only empirically. The first pairs of relations (1) can be considered as *parametric representations* of relations between N_z and r_z , where z is called a *parameter*. The third relations represent the definition of the length. If we eliminate this parameter (z) from the first pairs of equations and take $r_0 = 1$ for clarity, we obtain the relation $N_z = f(r_z)$ in the form

$$N_z = P \cdot r_z^{-D}, \quad (2)$$

where D can be thought of as the *fractal dimension* and P is a constant prefactor. This inverse proportionality relationship between the number of segments N_z and the segment length r_z determines one of the basic laws of fractal geometry. It can be thought of as the *axiom of segment counting*.

Since the length of a prefractal is, by definition, $L_z = N_z \cdot r_z$, it follows from (2) that

$$L_z = P \cdot r_z^\alpha, \quad (3)$$

where α is the *scaling exponent*. This relationship determines also one of the fundamental laws in fractal theory known as the *power law*. It is to be noted that in geometry, there exist

notions such as “axiom”, “theorem”, “lemma” etc. Therefore, in fractal geometry, Eq. (3) represents a theorem. Both equations (2) and (3) are revealed as straight-line plots when the results of counting the number of segments and that of calculating the value of lengths of a fractal are plotted on log-log axes against the values of the segment lengths at which they are estimated (Fig. 1C).

B. Geometrical and statistical self-similarity

The object's property known as *self-similarity* was first invented in [7] and can be geometrical or statistical [3,6,10,19]. A fractal pattern is said to be *geometrically self-similar* if each small piece of it is duplicate of the whole object [3]. Nevertheless, this definition should be quantified since small pieces that constitute geometrical or natural objects are rarely identical copies of the whole object [20].

A more exact interpretation of this descriptive definition have been offered in [4] introducing a *generating element* of a generator, which is usually made up of straight-line segments. A particular and logical concatenation of some segments of a generator could be thought of as the generating element of a generator if the whole object can be completely built with such elements by their translations and/or rotations.

Two generating elements of two generators of a fractal set can be geometrically similar or not. According to the definition of similarity in Euclidean planimetry two generating elements of the generators at stages z and $z + 1$ are similar to each other if (a) the ratio of the measure of a segment of the generating element at stage $z + 1$ and the measure of the corresponding segment of the element at stage z is constant for all pairs of corresponding segments, and (b) the angles between the pairs of corresponding segments of the two generating elements are congruent [4].

The pieces of natural objects, mostly the pieces of biological objects [6] are rarely exact reduced copies of the whole object; rather than being geometrically self-similar, they are *statistically self-similar*. It is stated that the natural object is statistically self-similar if a property of every small piece of an object is not significantly different from same property measured on the whole object [2,3,6]. A pattern in nature is hypothesized to be a fractal-like, i.e., statistically self-similar across a range of scales [1]. Some authors tried to graphically illustrate the self-similarity of geometrical and natural fractals [6,9,12] but these illustrations were not in full accordance with the above mentioned definition. In fact, a natural object can be a statistical fractal if (a) it is statistically self-similar to a complex system whose this object is a part, or (b) it consists of many small pieces being statistically self-similar to the whole object.

C. Fractal analysis

Previous work on fractal analysis represented traditionally a two-dimensional analysis directed mainly to the analysis of border's outlines of natural objects [2]. Fractal analysis is a mathematical and experimental discipline which uses methodology of Mandelbrot's fractal geometry and Euclid's traditional geometry to investigate the structure function and other properties of real objects. The problem is

directed to calculating the fractal dimension and analyzing what the value of this parameter tells us about the considered object.

By analogy with investigation of a geometrical fractal set in fractal geometry, the use of fractal analysis signifies specifying four things: (1) the shape of a starting object (which corresponds to the initiator of a fractal set), (2) algorithm, enabling its repeated application to the starting object, (3) conditions (before all, the property of *scaling*) and (4) fractal dimension. The *limit generator* is the same as the starting object, because it was not changed during repeated application of the algorithm [4].

In fractal analysis there are two basic approaches to measuring the fractal dimension of objects in a plane [2]. The first and most commonly used is *length-related method* comprising the classical Richardson's coastline method [7], box-counting method and dilation method [21]. The second method is *mass-related method* [2]. The use of different methods of fractal analysis has made comparison of results difficult because each method of determination of the fractal dimension gives slightly different results when uses to analyze the same structure [3,11].

D. Length-related methods

Lewis F. Richardson wondered if there are more wars between nations that share longer common borders [22]. Answering this question required measuring the length of national borders. To measure the length of the border drawn on a map, he used as a ruler a divider having sharp points at the end of both arms. The total length of the border was given by the number of divider steps multiplied by the distance between the ends of the divider.

When he reduced the distance between the ends of the divider and repeated the entire measurement, the length did not converge to a stable value but kept increasing. That is, the length *scales* with the resolution of the instrument used to measure such a length. Empirical evidence suggests that the smaller the increment of measurement, the longer the measured length becomes. It seems that the measured length increased without limit as the measurement scale decreased towards zero. He concluded that lengths of the coastlines and frontiers between countries can be given by

$$L(r) \propto F \cdot r^{1-D}, \quad (4)$$

where the value of the exponent D seems to depend upon the coastline that is chosen. Topology fails to discriminate between different coastlines [11]. To Richardson, the D was a simple exponent of no particular significance.

Having “unearthed” Richardson's work in which he claimed that his lines' slopes had no theoretical interpretation [11], Mandelbrot proposed that the exponent D should be interpreted as a fractal dimension [7]. It is our impression that the Richardson's work had a primary and decisive influence on development of Mandelbrot's fractal geometry. Also, we find that among many fractal analysis techniques, only the Richardson's ruler-counting method enables calculation of the length of an object's border or irregular

line. The other techniques analyze the slopes of the straight lines representing the relationship between the number of scale steps and scale length on log-log axes. However, it should bear in mind that the *number* of distinct scales of length of natural patterns is, for all practical purposes, always infinite [3,11].

Segment-counting method of measuring the fractal dimension is robust with very high correlation coefficients but it is, at the same time, tedious and time consuming. Therefore, the need for more handsome methods for that measurement emerges. The conventional box-counting [2] is one method suitable to measure fractal dimensions of real objects [2-4,23,24].

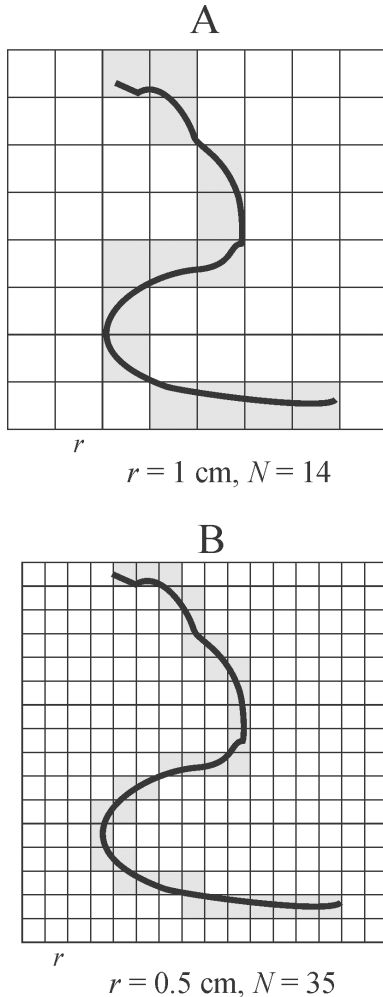


Fig. 2. Application of the box-counting method to unformatted curve. (A) The image is covered with a set of larger squares and the squares which cover the curve (gray squares) are counted. (B) The image is covered with a set of smaller squares: the squares which cover the curve are counted.

This method is based on the concept of “covering” the image with rectangular coordinate grid [21]: the image (e.g., the curve) is overlaid with a grid, and the number of boxes intersected by the pattern counted. Each set of boxes is characterized by the square side (r). The corresponding number of squares (N) necessary to cover the border is

presented as a function of r . Thus, $N(r)$ is the number of boxes that contain at least one point of the object for different box sizes. Fractal dimension D of an image is determined as the slope of the log-log relationship between $N(r)$ and r . Although the method is not devoted to measuring the lengths and other features of curves [25], this is the best technique to estimate the fractal dimension. It follows from the fact that the number of boxes covering the image is inversely related to the square side (Fig. 2).

The ideas underlying the fractal dimension are related to the choice of “balls” [6]. In the space which has one dimension the ball is a straight-line segment, in two dimensions space the ball is a circle etc. The selection of squares (instead of circles) as “balls” is customary since it can readily be implemented on a computer. Usually all transformations and measurements were carried out on a computer using the public domain Java image processing program ImageJ (www.rsb.info.nih.gov/ij/) developed at the US National Institute of Health. Therefore the box-counting method, which is based on this idea, has been shown to be most powerful and commonly used among other fractal techniques [3-5,24,26].

Unfortunately, it seems that the box-counting method demonstrates some errors. One of the main shortcomings of the box-counting method is that for large boxes the number of boxes intersected is sensitive to the location of the object upon the grid [2], so it is necessary to average many trials (maybe 200) for each grid size [10,11]. It seems that dilation method, based on the Minkowski-Bouligand dimension [7], is less sensitive to the location of the image in a frame [2]. This method replaced each pixel of the border by a circle whose diameter ranged from 3 to 61 pixels [3].

III. MATERIALS AND METHODS

A. Image processing

Golgi impregnated neurons of the human and rat spinal cords were traced using camera lucida [27]. The images were obtained from 30 human and 23 rat cells. Of these, 23 images of human neurons were previously published [28], while previously unpublished images (7 human neurons and 23 rat neurons) were taken from [29] with permission. A detailed description of the histological procedure can be found in [27,28].

The drawings of neurons were converted into digitized images using a scanner (“Optic Pro 9630P”, Mustek) with a resolution of 600 dpi. To avoid the problems associated with large measuring elements [30], we ensured that the digitized images, printed on paper (with the same resolution) by a printer (Hewlett-Packard 5L), were no larger than A5 size (i.e., 15 x 21 cm). Further transformations were carried out using the Image J software. All scanned images (Fig. 3A) were imported: using the corresponding tools, axons, spines (if present) and soma were removed digitally from the drawing. After that, each dendrite was filled with pixels (Fig. 3B). Then, the program performed a skeleton of the image to a stick figure (Fig. 3C). In such a computer-generated image each dendrite was presented by a curve whose thickness was one pixel.

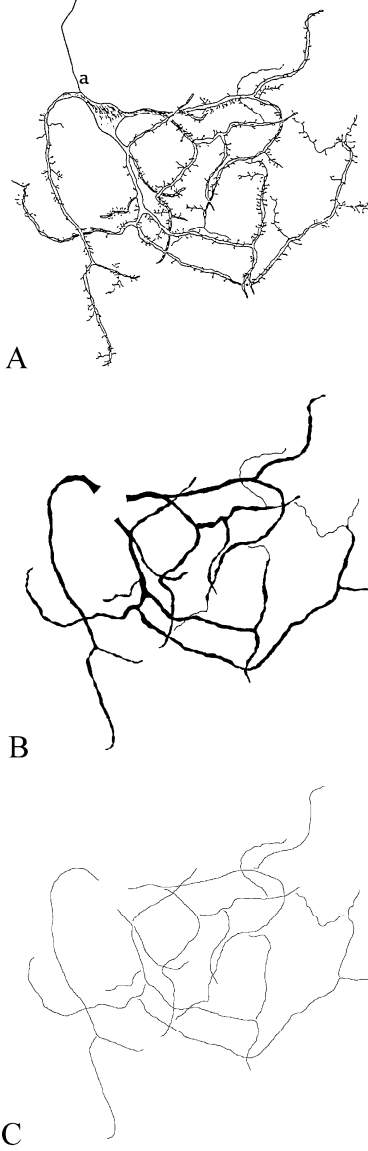


Fig. 3. A drawing of neurons is converted into digital image (A) (from [28] – with permission) and imported into the “Image J” software. Axons, spines and soma are digitally removed and each dendrite is filled with pixels (B). The software has produced a skeleton of the image (C).

B. Methods of image analysis

Skeletonized images, plotted on paper, were analyzed manually, using the ruler-counting method [25]. This method is based upon classical “coastline method” of Richardson involving the measurement of the length of a dendrite with rulers of various lengths [4].

For that purpose we employed a divider keeping a fixed distance between the two divider ends during each measurement. Measurements of the number of divider steps (N) and the dendritic length (L) were done around a “skeletonized” curve in the image. For a chosen ruler length r we counted out the steps (N) of the divider first, and then we obtained the scaled length (L) of the dendrite as the number

of divider steps multiplied by the distance between the divider ends. Measurements of all dendrites were carried out using the ruler spans between 2 mm and up to 25 mm. The approximate total length of all neuronal dendrites was the sum of lengths of each dendrite and dendritic branch. Since the total length depends from the number of divider steps (see Eq. (4)), the global fractal dimension (D) of each neurons was calculated.

In the next step, each image was subjected to the box-count method, which is implemented in Image J software, in order to calculate box dimension (D_B). For detailed explanation of box-counting procedure the reader is referred to [31,32]. The box sizes for the box-counting method were taken from 2^1 to 2^k pixel, where k is the value for which N is equal to unity.

C. Statistics

We have applied a one-way ANOVA to the mean fractal dimensions of different laminae to investigate whether there is statistical significance to the differences between their means. For cases where one-way ANOVA has shown statistically significant differences between sample means, we have adopted the Scheffé post-hoc test for matched pairs [33].

IV. RESULTS

Table I shows the results of two methods of fractal analysis applied to images of the human neurons in first four laminae. All mean values of fractal dimension obtained applying the ruler-counting method is smaller than those obtained for box-counting method. When applying ruler-counting method, Scheffé’s post-hoc test indicated a significant difference between laminae IV and all other three laminae ($p < 0.05$). Then again, when box-count method is applied to the same images, a significant difference is found between the following pairs of data: laminae I-II, laminae I-IV and laminae III-IV ($p < 0.05$). Further, first method of fractal analysis presented difference between laminae IV and others, while second method showed difference between three laminae of the human spinal cord.

Table I. The cell number, values of fractal dimensions and difference between two fractal dimension (δ) of the human neurons in first four laminae of the spinal cord. Each value is presented as average $\pm$ standard error (AV $\pm$ SE).

Lamina	Fractal dimension		$\delta(\%)$
	D	D_B	
I	1.06 ± 0.03	1.08 ± 0.01	0.9-5.4
II	1.04 ± 0.05	1.13 ± 0.01	1.2-8.3
III	1.07 ± 0.03	1.11 ± 0.01	0.7-6.6
IV	1.11 ± 0.02	1.15 ± 0.01	2.7-4.5

The results of applying two methods of the fractal analysis to skeletonized images of rat neurons are shown in Table II. Similarly, all mean values in first four laminae of global fractal dimension are smaller than those obtained for box-counting method. In contrast, an opposite conclusion can be drawn for neurons in laminae V and VI. For global fractal dimension, significant difference was obtained between the following pairs: laminae I-VI and laminae II, III-VI ($p <$

0.05). In the case of box-dimension, a significant difference was obtained between laminae I-VI, laminae II, III-VI and laminae IV-VI ($p < 0.05$). Unlike ruler-counting method, which found difference in three laminae, box-count method differentiated four laminae of the rat spinal cord. In both cases (the human and rat spinal cord), the difference between two methods of fractal analysis varied up to 10% (Tabs. I and II).

Table II. The cell number, values of fractal dimensions and difference between two fractal dimension (δ) of the rat neurons in first six laminae of the spinal cord. Each value is presented as average $\pm$ standard error (AV $\pm$ SE).

Lamina	Fractal dimension		$\delta(\%)$
	D	D_B	
I	1.07 ± 0.02	1.07 ± 0.01	0.3-6.4
II-III	1.06 ± 0.01	1.10 ± 0.02	0.5-4.5
IV	1.05 ± 0.02	1.09 ± 0.01	0.6-5.6
V	1.04 ± 0.02	1.03 ± 0.03	3.3-8.7
VI	1.044 ± 0.006	1.03 ± 0.03	2.1-7.3

V. DISCUSSION

A. Fractal theory and systems in the nature

Various systems in nature, if they accomplish four properties (Subsection IIA) during their construction or formation can be thought as fractals. Like we said before, geometrical self-similarity means that every small piece of an object resembles the whole object (Subsection IIB). Many authors tried to demonstrate graphically this definition [1,6,12]. We showed that such two objects can be self-similar if their generating elements are geometrically similar [4]. Any two prefractals of the Koch set are not similar, contrary to their generating elements. This is obvious from Fig. 1A, middle and right side: these two images could only be visually alike, but not geometrically similar.

The simplest scaling relationship between some property of the system and the scale has the power law form [6], or precisely an inverse power law scaling. It should be noted that this property is *necessary* but not *necessary and sufficient* for the fractality of a set of objects [4]. In order that a set of objects represents a fractal set, it is *necessary* that this set satisfies the *inverse power law scaling*. As has already been mentioned, the perimeter of mathematical prefractals tends to infinity when the segment length of the curves decrease [Subsection IIA).

However, where one analyzes the fractal properties of real objects, this law should change its form. For example, as the lengths of the polygonal segments decrease and finer details of the structure are revealed, the measured lengths of the polygons increase converging to an asymptotically stable value. Generally, the power law scaling can be thought of as the *polynomial law scaling* [17]. Such law is characteristic of geometrical and real objects in fractal analysis. In order to calculate the fractal dimension of the circle as a non-fractal object, Ristanovic et al. [17] presented the graph of relationship between the number of polygonal segments and their sizes on log-log axes. It is seen that the obtained data can be fitted with a decreasing straight line with the fractal

dimension $D = 1.008$. Although the circle and the set of polygons are not a fractal set yet they have the corresponding fractal dimension. Thus, it is conclusion of previous finding, that measurement of the fractal dimension is usually not intended to indicate whether an object is a fractal object or not [3].

B. Segment-count versus box-count in fractal analysis

In this report, we have compared two different fractal methods to calculating fractal dimension of skeletonized images of neurons over different laminae in the human and rat spinal cord. The results show that a significant difference in the mean fractal dimension between groups of cells from different laminae exists in both methods. But, more pairs were found when the box-counting method is applied. One may consider that results shown in this paper are consequential to the low count of neurons included in the study. We cannot accept this attitude, since we have chosen only images of the neurons that were characteristic for each particular laminae.

All in all, present results are in accordance with our previous paper where we have showed that box-count method, in the case of spinal cord neurons, is the most sensitive than other methods of fractal analysis [24]. Moreover, the box-count method is based on the idea of covering the whole image with sets of square boxes, which is in agreement with the procedure of the calculation of fractal dimension for an object as originally suggested by Mandelbrot [7].

The traditional Richardson's segment-count method is by no means basic for measuring the fractal dimension within segment-related techniques. Unfortunately, this measuring is tedious. It would be best to prepare a suitable program for that analysis. In our previous paper, we have offered the *circle-count method* [17]. This is a technique for measuring the length of cell dendrites. The analytical image was input into our proprietary software for drawing circles, written in Visual Basic (Microsoft) and the lengths of dendrites were successfully measured. Unfortunately, segment-count method, or any adaptation of this technique, cannot measure or quantify space-filling property of an object.

Alternatively, when the box-count method, is applied to neuronal images, the fractal dimension (i.e. the box dimension) depends on image presentation. While the fractal dimension of outline images defines the irregularity in the shape of the image [32], the same parameter for skeletonized images summarizes the degree of dendrite aberrations from straight lines [34]. Our previous study present an explanation how the box-count method can estimate the area of the binary image of the neuron, or how the area of the neuron deviates from the area of square which encompassed the neuron [35].

The fractal dimension is the basic parameter in fractal theory. It is a measure of pattern's complexity [30], precisely this parameter analyzes how details of a pattern change with the scale at which they are measured or it estimates the degree of tortuosity and irregularity of smooth lines comprising a pattern [3,4]. It is our impression that box-count

method only can fulfill object's space-filling property. We have shown that this conclusion is true in case of two dimensional images of the neuron [35].

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